

Hw 7

Keegan Burr

11/25/2024

Recall that in class we showed that for randomized response differential privacy based on a fair coin (that is a coin that lands heads up with probability 0.5), the estimated proportion of incriminating observations $\hat{P}$ ¹ was given by $\hat{P} = 2\hat{\pi} - \frac{1}{2}$ where $\hat{\pi}$ is the proportion of people answering affirmative to the incriminating question.

I want you to generalize this result for a potentially biased coin. That is, for a differentially private mechanism that uses a coin landing heads up with probability $0 \leq \theta \leq 1$, find an estimate $\hat{P}$ for the proportion of incriminating observations. This expression should be in terms of θ and $\hat{\pi}$.

$$\begin{aligned}\hat{\pi} &= \theta(1-\theta) + \theta P \\ \theta P &= \hat{\pi} - \theta(1-\theta) \\ P &= (\hat{\pi} - \theta(1-\theta))/\theta\end{aligned}$$

Next, show that this expression reduces to our result from class in the special case where $\theta = \frac{1}{2}$.

$$\begin{aligned}P &= (\hat{\pi} - 1/2(1-1/2))/1/2 \\ P &= (\hat{\pi} - 1/2(1/2))/1/2 \\ P &= (\hat{\pi} - 1/4)/1/2 \\ P &= (2\hat{\pi} - 1/2)\end{aligned}$$

Part of having an explainable model is being able to implement the algorithm from scratch. Let's try and do this with KNN. Write a function entitled `chebychev` that takes in two vectors and outputs the Chebychev or L^∞ distance between said vectors. I will test your function on two vectors below. Then, write a `nearest_neighbors` function that finds the user specified k nearest neighbors according to a user specified distance function (in this case L^∞) to a user specified data point observation.

```
#student input
```

```
#chebychev function
```

¹in class this was the estimated proportion of students having actually cheated

```

chebychev <- function(x, y) {
  return(max(abs(x - y)))}

#nearest_neighbors function (Taken from notes)
nearest_neighbors = function(x, obs, k, dist_func){
  dist = apply(x, 1, dist_func, obs) #apply along the rows
  distances = sort(dist)[1:k]
  neighbor_list = which(dist %in% sort(dist)[1:k])
  return(list(neighbor_list, distances))
}

x<- c(3,4,5)
y<-c(7,10,1)
# Changed from cheby to chebychev so no error produced
chebychev(x,y)

```

Finally create a `knn_classifier` function that takes the nearest neighbors specified from the above functions and assigns a class label based on the mode class label within these nearest neighbors. I will then test your functions by finding the five nearest neighbors to the very last observation in the `iris` dataset according to the `chebychev` distance and classifying this function accordingly.

```

library(class)
df <- data(iris)
#student input (Taken from notes)
knn_classifier = function(x,y){

  groups = table(x[,y])
  pred = groups[groups == max(groups)]
  return(pred)
}

#data less last observation
x = iris[1:(nrow(iris)-1),]
#observation to be classified
obs = iris[nrow(iris),]

#find nearest neighbors
ind = nearest_neighbors(x[,1:4], obs[,1:4],5, chebychev)[[1]]
as.matrix(x[ind,1:4])
obs[,1:4]
knn_classifier(x[ind,], 'Species')
obs[, 'Species']

```

Interpret this output. Did you get the correct classification? Also, if you specified $K = 5$, why do you have 7 observations included in the output dataframe?

Given our iris dataset and based on our knn function, our observation belongs to the virginica class. This is the correct classification because the majority of nearest neighbors belonged to the virginica class. In regards to why we have 7 observations in the dataframe instead of 5, this is because there seems to be a couple of datapoints that were tied in the distance. You can see points 102 and 143 is one of the examples that were tied.

Earlier in this unit we learned about Google's DeepMind assisting in the management of acute kidney injury. Assistance in the health care sector is always welcome, particularly if it benefits the well-being of the patient. Even so, algorithmic assistance necessitates the acquisition and retention of sensitive health care data. With this in mind, who should be privy to this sensitive information? In particular, is data transfer allowed if the company managing the software is subsumed? Should the data be made available to insurance companies who could use this to better calibrate their actuarial risk but also deny care? Stake a position and defend it using principles discussed from the class.

I believe sensitive healthcare data, like the data managed by Google's DeepMind, should not be shared with insurance companies without consent that is fully explicit and informed. According to J.S. Mill's Harm Principle, people should be protected from potential harm and in this case, allowing insurance companies to access this kind of data could lead to major discrimination or an individual being denied insurance which can be extremely harmful. Overall, I believe patients should retain control over their data. If the data is going to be transferred, consent or re-consent should be required because it is important to respect the autonomy of the patients.

I have described our responsibility to proper interpretation as an *obligation* or *duty*. How might a Kantian Deontologist defend such a claim?

A Kantian Deontologist would defend the claim based on Kant's formulations. First, with the idea to treat individuals as ends, never merely as means to an end, proper interpretation is part of recognizing the autonomy of individuals, ensuring that they are understood. Disregarding this interpretation undermines their autonomy treating them merely as means. Then by making sure the claim is universalizable, if we expected others to give us a fair interpretation, that means we must also apply the same standard to ourselves and give a fair interpretation to others.